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Phase 3: Motorcycle Dashboard

Reflection Questions

1. How was non-blocking input implemented?

Non-blocking input was implemented using terminal raw mode with `termios`, `read()`, and `fcntl()` so the program could detect key presses without pausing the dashboard or other running threads. This allowed the user to press controls like W, K, I, and Q while the motorcycle simulation continued updating in real time.

2. How does the input thread interact with other subsystems?

The input thread interacts with the other subsystems by updating shared variables inside `global_state`, such as engine status, speed mode, lights, refueling status, battery mode, and shutdown status. It uses mutex locks before changing shared data so that the engine, motion, fuel, hybrid, ECU, dashboard, and logger threads do not access the same values unsafely.

3. How does the ECU handle invalid user actions?

The ECU handles invalid user actions by checking the current system state and enforcing rules instead of letting input directly override the system. For example, it limits speed during low fuel, caps RPM and speed during overheating, keeps RPM at idle when stopped, and updates system zones like fuel status, temperature status, and RPM status.

4. What challenges were encountered when integrating input with synchronization?

One challenge was making sure threads did not lock each other out while reading or updating shared state. For example, when the hybrid thread continued while the engine was off without unlocking its mutexes, the ignition key could not turn the engine back on. This was fixed by making sure every lock was released before continuing or leaving a thread loop.

5. How was the graceful shutdown implemented?

Graceful shutdown was implemented using the `shutdown_flag`. When the user presses Q, the input thread sets this flag, broadcasts condition variables to wake waiting threads, restores the terminal settings, and allows each thread loop to exit safely before the program joins the threads and destroys the mutexes.